

CATS⁺

Competishun Advance

Training Sessions

(Lecture - IX)

Column Matching

Column-I

Column-II

a. If $\lim_{x \rightarrow \infty} (\sqrt{x^2 - x - 1} - ax - b) = 0$, where $a > 0$, then there exists at least one a and b for which point $(a, 2b)$ lies on the line.	p. $y = -3$
b. If $\lim_{x \rightarrow 0} \frac{(1 + a^3) + 8e^{1/x}}{1 + (1 - b^3)e^{1/x}} = 2$, then there exists at least one a and b for which point (a, b^3) lies on the line.	q. $3x - 2y - 5 = 0$
c. If $\lim_{x \rightarrow \infty} (\sqrt{x^4 - x^2 + 1} - ax^2 - b) = 0$, then there exists at least one a and b for which point $(a, -4b)$ lies on the line.	r. $15x - 2y - 11 = 0$
d. If $\lim_{x \rightarrow -a} \frac{x^7 + a^7}{x + a} = 7$, where $a < 0$, then there exists at least one a for which point $(-a, 2)$ lies on the line.	s. $y = 2$

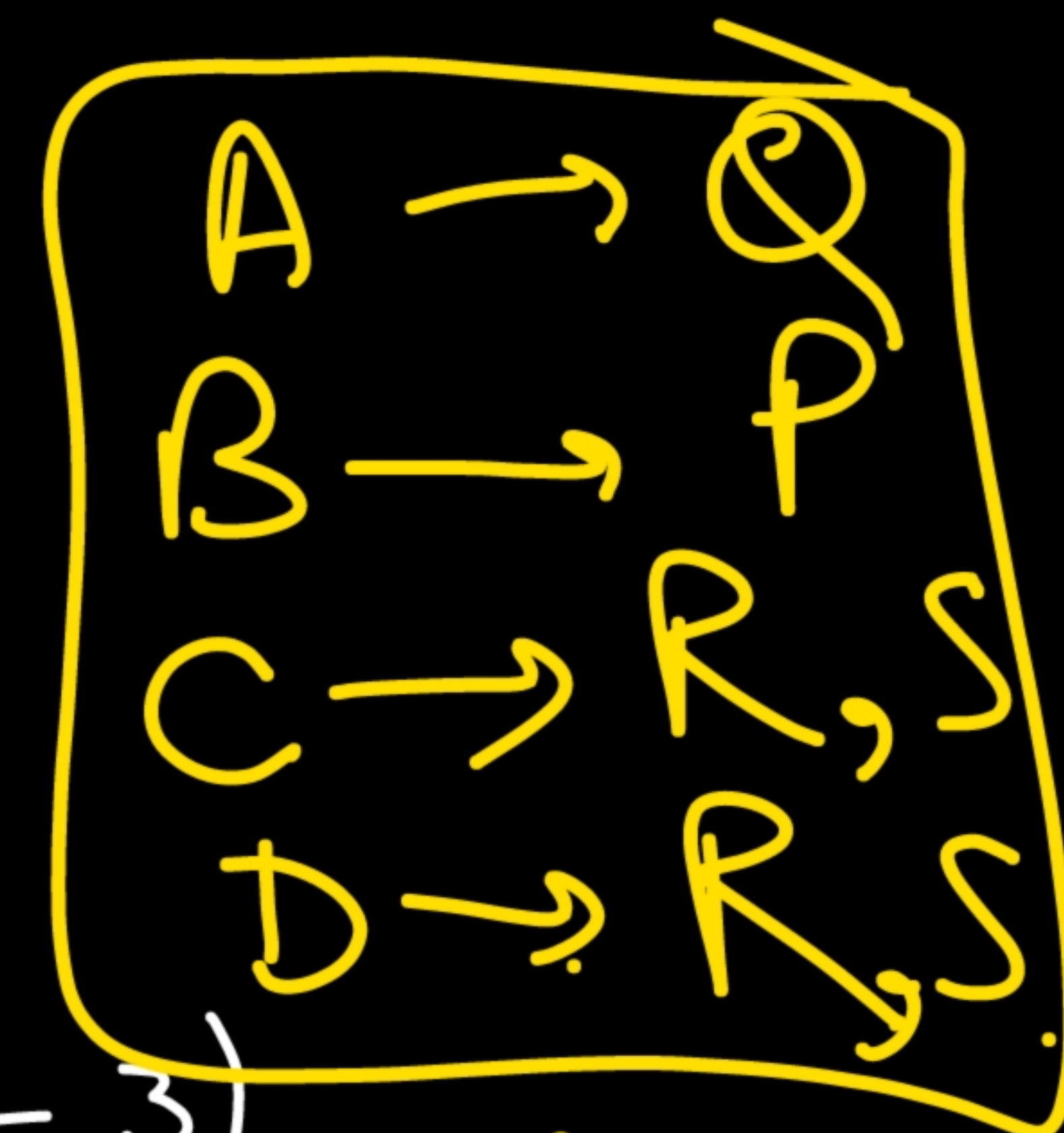
A $(a, b) \equiv (1, -\frac{1}{2})$
 Q $(a, b) \equiv (1, -1)$

B $(a, b) \equiv (1, -3)$

C $(a, b) \equiv (1, -\frac{1}{2})$

$(a, -4b) \equiv (1, 2)$

D $(-a, 2) \equiv (1, 2)$



A

$$\textcircled{c} \lim_{x \rightarrow 0} \left(\sqrt{x^4 - x^2 + 1} - ax^2 - b \right) = 0$$

$$\begin{array}{l} x \rightarrow -x \\ x \rightarrow a \end{array} \left[\begin{array}{l} \lim_{x \rightarrow -a} \frac{x^7 + a^7}{x + a} = 7 \\ \lim_{x \rightarrow a} \frac{x^7 - a^7}{x - a} = 7 \cdot a^{n-1} \end{array} \right] \Rightarrow \begin{array}{l} a^6 = 1 \\ a = \pm 1 \end{array}$$

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n \cdot a^{n-1}$$

$$\lim_{x \rightarrow 0} \frac{1 + a^3 + \underline{8 \cdot e^{1/x}}}{1 + (1-b^3) \underline{e^{1/x}}} = 2.$$

R.H.L.
 $x \rightarrow 0^+$
 $\frac{1}{x} \rightarrow \infty$
 $e^{1/x} \rightarrow \infty$

$$\frac{8 \cdot e^{1/x}}{(1-b^3) e^{1/x}} = 2$$

$$4 = 1 - b^3$$

$$\boxed{b^3 = -3}$$

L.H.L.
 $x \rightarrow 0^-$
 $\frac{1}{x} \rightarrow -\infty$
 $e^{1/x} \rightarrow 0$

$$\frac{1+a^3}{1} = 2$$

$$a^3 = 1 \Rightarrow \boxed{a=1}$$

(A)
 $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 - x - 1} - ax - b \right) = 0$

$$= x \left(1 - \frac{1}{x} - \frac{1}{x^2} \right)^{\frac{1}{2}} - ax - b$$

$$= x \left(1 + \frac{1}{2} \left(-\frac{1}{x} - \frac{1}{x^2} \right) \right) - ax - b$$

$$= (1-a)x - \frac{1}{2} - \frac{1}{2x} - b = \underline{\underline{0}}$$

$$1-a=0 \Rightarrow a=1$$

$$\textcircled{A} \quad -\frac{1}{2} - b = 0 \Rightarrow b = -\frac{1}{2}$$

Paragraph

Let $y = f(x)$ satisfies the equation

$$f(x) = (e^{-x} + e^x) \cos x - 2x - \int_0^x (x-t) f'(t) dt$$

Q-1 y satisfies the differential equation

☒ (1) $\frac{dy}{dx} + y = e^x (\cos x - \sin x) - e^{-x} (\cos x + \sin x)$

(2) $\frac{dy}{dx} - y = e^x (\cos x - \sin x) + e^{-x} (\cos x + \sin x)$

(3) $\frac{dy}{dx} + y = e^x (\cos x + \sin x) - e^{-x} (\cos x - \sin x)$

(4) $\frac{dy}{dx} - y = e^x (\cos x - \sin x) + e^{-x} (\cos x - \sin x)$

Q-2 The value of $f(0) + f'(0)$ equal

- (1) -1 ☒ (2) 0 (3) 1 (4) 2

Q-3 $f(x) =$

(1) $e^{-x} (\cos x - \sin x) + \frac{e^x}{5} (3 \cos x + \sin x) + \frac{2}{5} e^{-x}$

(2) $e^{-x} (\cos x + \sin x) + \frac{e^x}{5} (3 \cos x - \sin x) - \frac{2}{5} e^{-x}$

☒ (3) $e^{-x} (\cos x - \sin x) + \frac{e^x}{5} (3 \cos x - \sin x) + \frac{2}{5} e^{-x}$

(4) $e^{-x} (\cos x + \sin x) + \frac{e^x}{5} (3 \cos x - \sin x) - \frac{2}{5} e^{-x}$

$$f(x) = (e^{-x} + e^x) \cos(x) - 2x - \int_0^x (x-t) \cdot f'(t) \cdot dt$$

$$\begin{aligned} x=0 \rightarrow f(0) &= 2 \cdot 1 - 0 - \int_0^0 \dots \\ f(0) &= 2 \end{aligned}$$

$$= x \int_0^x f'(t) dt - \int_0^x t f'(t) \cdot dt$$

$$= x \left(f(t) \right)_0^x$$

$$= x(f(x) - f(0)) = x f(x) - 2x$$

$$\Rightarrow f(x) = (e^x + e^{-x}) \cos(x) - 2x - x f(x) + 2x + \int_0^x t f'(t) \cdot dt$$

$$f(x) = (e^x + e^{-x}) \cos(x) - x f(x) + \int_0^x t f'(t) \cdot dt$$

$$\frac{d}{dx} \rightarrow f'(x) = (e^x - e^{-x}) \cos(x) - \sin(x) (e^x + e^{-x}) - f(x) - x f'(x) + x f'(x)$$

$$f(x) + f'(x) = (e^x - e^{-x}) \cdot \cos(x) - (e^x + e^{-x}) \cdot \sin(x)$$

$$f'(x) + f(x) = \underline{e^x} (\underline{\cos(x)} - \underline{\sin(x)}) - \underline{e^{-x}} (\underline{\cos(x)} + \underline{\sin(x)})$$

$$\underbrace{x=0}_{\text{I.F.}} \rightarrow f'(0) + f(0) = 1 - 1 = 0$$

$$\text{I.F.} = e^x \rightarrow e^x \cdot f'(x) + e^x \cdot f(x) = e^{2x} (c - x) - (c + x)$$

$$\Rightarrow \frac{d}{dx} (f(x) \cdot e^x) = e^{2x} (c - x) - (c + x)$$

$$\int \rightarrow f(x) \cdot e^x = \int e^{2x} (c - x) dx - (\sin(x) - \cos(x)) + C$$

Let $I_1 = \int e^{2x} (\cos(x) - \sin(x)) dx$

$$= \underbrace{\frac{e^{2x}}{2} (c - s)}_{\frac{5}{4} I_1} - \underbrace{\frac{e^{2x}}{4} (-s - c)}_{-I_1} - \int \frac{e^{2x}}{4} (-c + s) dx$$

$$\frac{5}{4} I_1 = \frac{e^{2x}}{4} (2c - 2s + s + c)$$

$$I_1 = \frac{e^{2x}}{5} (\underline{3c - s})$$

$$I_1 = \frac{e^{2x}}{2} (c - s) + \int \frac{e^{2x}}{2} (+s + c) dx = \underbrace{\frac{e^{2x}}{2} (c - s)}_{I} + \frac{1}{2} \left(\frac{e^{2x}}{2} (s + c) - \int \frac{e^{2x}}{2} (c - s) dx \right)$$

$$f(x) \cdot e^x = I_1 - (5 - c) + C$$

$$f(x) e^x = \frac{e^{2x}}{5} (3c - 8) - (8 - c) + C$$

$$\div e^x \left[f(x) = \frac{e^x}{5} (3 \cos(x) - \sin(x)) - e^{-x} (\sin(x) - \cos(x)) + C \cdot e^{-x} \right]$$

$$\left[x=0 \right] 2 = \frac{3}{5} + 1 + C \Rightarrow C = 1 - \frac{3}{5} = \frac{2}{5}$$

$$f(x) = \frac{e^x}{5} (3 \cos(x) - \sin(x)) - e^{-x} (\sin(x) - \cos(x)) + \frac{2}{5} e^{-x}$$

Paragraph

Let the equation $\sqrt{x - \frac{1}{x}} + \sqrt{1 - \frac{1}{x}} = x$, has a solution

$\frac{a + \sqrt{b}}{c}$, where a, b, c are coprime natural numbers, then answer the

following:-

Q-1 Total number of triplets (a, b, c) are
☐ A 0 ☒ B 1 ☐ C 2 ☐ D 3

Q-2 The value of ' $a + b + c$ ' is
☐ A Prime Number ☒ B Composite Number

☐ C Odd Number ☐ D None of A, B, C

$$\frac{a + \sqrt{b}}{c} = \frac{1 + \sqrt{5}}{2}$$

$$a + b + c = 8$$

$$\sqrt{x - \frac{1}{x}} + \sqrt{1 - \frac{1}{x}} = x \longrightarrow \sqrt{1 - \frac{1}{x}} = x - \sqrt{x - \frac{1}{x}}$$

$\Downarrow ()^2$

$$1 - \frac{1}{x} = x^2 + x - \frac{1}{x} - 2 \cdot x \cdot \sqrt{x - \frac{1}{x}}$$

$$\begin{aligned} (1)^2 \left[\begin{aligned} \sqrt{x - \frac{1}{x}} &= x - \sqrt{1 - \frac{1}{x}} \\ \Rightarrow \cancel{x - \frac{1}{x}} &= x^2 + \cancel{1 - \frac{1}{x}} - 2x \cdot \sqrt{1 - \frac{1}{x}} \\ x^2 - x + 1 - 2x \cdot \sqrt{1 - \frac{1}{x}} &= 0 \end{aligned} \right. \end{aligned}$$

$$\begin{aligned} \div x \left[\begin{aligned} x^2 + \underline{x} - 1 - 2 \cdot x \cdot \sqrt{x - \frac{1}{x}} &= 0 \\ \Rightarrow \left(x - \frac{1}{x}\right) + 1 - 2 \cdot 1 \cdot \sqrt{x - \frac{1}{x}} &= 0 \\ \left(\sqrt{x - \frac{1}{x}} - 1\right)^2 = 0 &\Rightarrow x - \frac{1}{x} = 1 \end{aligned} \right. \end{aligned}$$

$$x^2 - x - 1 = 0$$

$$x = \frac{1 \pm \sqrt{5}}{2}$$

Ques

If $y = \frac{(a+x)(b+x)}{(c+x)}$, where $a, b > c$ and $x > -c$, then which of the following is/are true

- (A) minimum value of y is $[\sqrt{a-c} + \sqrt{b-c}]^2$
 (B) minimum value of y occur at $x = \sqrt{(a-c)(b-c)} - c$
 (C) minimum value of y is $[\sqrt{a-c} - \sqrt{b-c}]^2$
 (D) None of these

$$y = \frac{(x+a)(x+b)}{x+c}, \quad a, b > c \quad \textcircled{8} \quad x > -c$$

$$\left[\begin{array}{l} x+c > 0 \\ \underline{t > 0} \end{array} \right]$$

$$y = \frac{(x+c+a-c)(x+c+b-c)}{x+c}$$

$$= \frac{(t+(a-c))(t+(b-c))}{t} = \frac{t^2 + (a+b-2c)t + (a-c)(b-c)}{t}$$

$$= (a+b-2c) + \frac{t}{t} + \frac{(a-c)(b-c)}{t} \geq (a-c) + (b-c) + 2\sqrt{(a-c)(b-c)} = \underbrace{(\sqrt{a-c} + \sqrt{b-c})^2}_{\text{Min. Value}}$$

$$\frac{\alpha + \beta}{2} \geq \sqrt{\alpha\beta}$$

$$\alpha + \beta \geq 2\sqrt{\alpha\beta}$$

for equality

$$t = \frac{(a-c)(b-c)}{t}$$

$$t = \sqrt{(a-c)(b-c)}$$

$$= x+c$$

$$x = \sqrt{(a-c)(b-c)} - c$$

Ques

A quadratic polynomial with leading coefficient unity is such that $P(x) = 0$ and $P(P(P(x))) = 0$ have a common root then which of the following is/are true

(A) $P(0) \cdot P(1) = 0$

(B) $P(0) \cdot P(1) < 0$

(C) $P(0) \cdot P(1) > 0$

(D) none of these

$$P(x) = x^2 + bx + c. \quad \xrightarrow{x=0} \quad P(0) = c.$$

$$P(x) = 0 \rightarrow \alpha, \beta$$

$$P(\alpha) = 0$$

$$\textcircled{2} P(\beta) = 0$$

$$\textcircled{1} P(P(P(x))) = 0 \rightarrow \alpha, \dots$$

$\Downarrow$

$$P(P(P(\alpha))) = 0.$$

$$P(P(0)) = 0.$$

$$P(c) = 0.$$

$$\Rightarrow c \cdot (c + b + 1) = 0.$$

$P(0) \cdot P(1) = 0$

$$P(c) = c^2 + bc + c = 0.$$

Paragraph If $a+b+c=4$, $a^2+b^2+c^2+3(ab+bc+ca)=21$, where $\underline{a, b, c \in \mathbb{R}}$
 then answer the followings:-

Q-1 The minimum value of $\underline{(a+b)} \cdot \underline{(b+c)} \cdot \underline{(c+a)}$ is
☐ A 3 ☒ B 18 ☐ C 2 ☐ D None of these.

Q-2 The maximum value of $(a+b)(b+c)(c+a)$ is
☐ A 3 ☐ B 18 ☐ C 2 ☒ D None of these

$$\begin{array}{l} \text{let } a+b=x \\ \quad b+c=y \\ \quad + c+a=z \\ \hline 2(a+b+c) = x+y+z \\ x+y+z=8 \end{array}$$

$$a^2+b^2+c^2+3(ab+bc+ca)=21$$

$$\Rightarrow (a^2+\Sigma ab) + (b^2+\Sigma ab) + (c^2+\Sigma ab) = 21$$

$$(a+b)(a+c) + (b+c)(b+a) + (a+c)(b+c) = 21$$

$$xz + xy + yz = 21$$

$$\begin{array}{l} (x+y) \cdot (x+y) \\ = x^2 + \underline{x(y+y)} + y^2 \end{array}$$

Given

$$x+y+z=8$$

$$xy+yz+zx=21$$

Now

$$(a+b)(b+c)(c+a) = x \cdot y \cdot z = \lambda \text{ (say)}$$

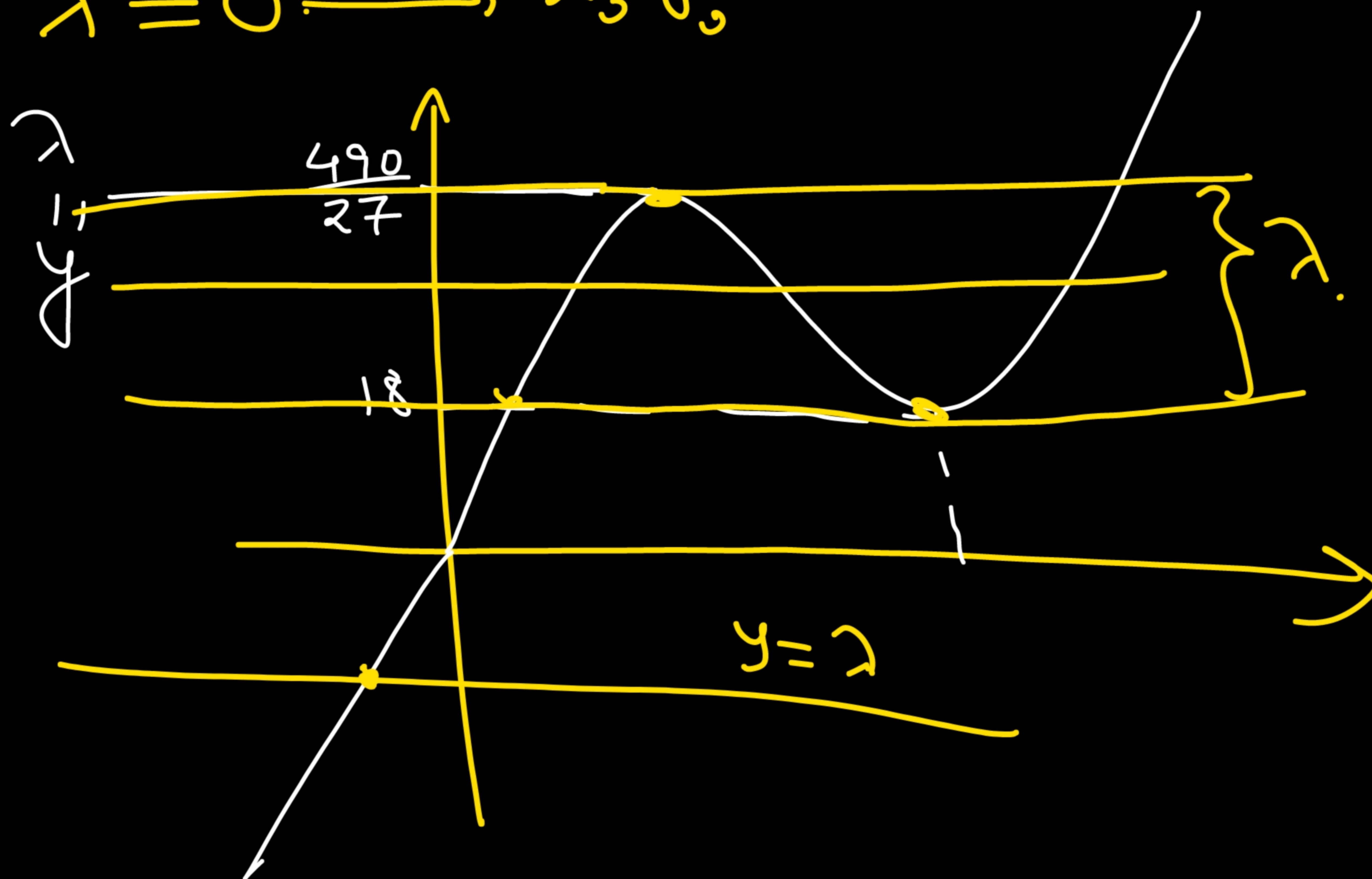
Consider a cubic equation

$$t^3 - 8t^2 + 21t - \lambda = 0 \longrightarrow x, y, z.$$

$$\Rightarrow t^3 - 8t^2 + 21t = \lambda$$

$y =$

$$\lambda \in \left[\underline{18}, \frac{490}{27} \right]$$



$$y = t^3 - 8t^2 + 21t$$

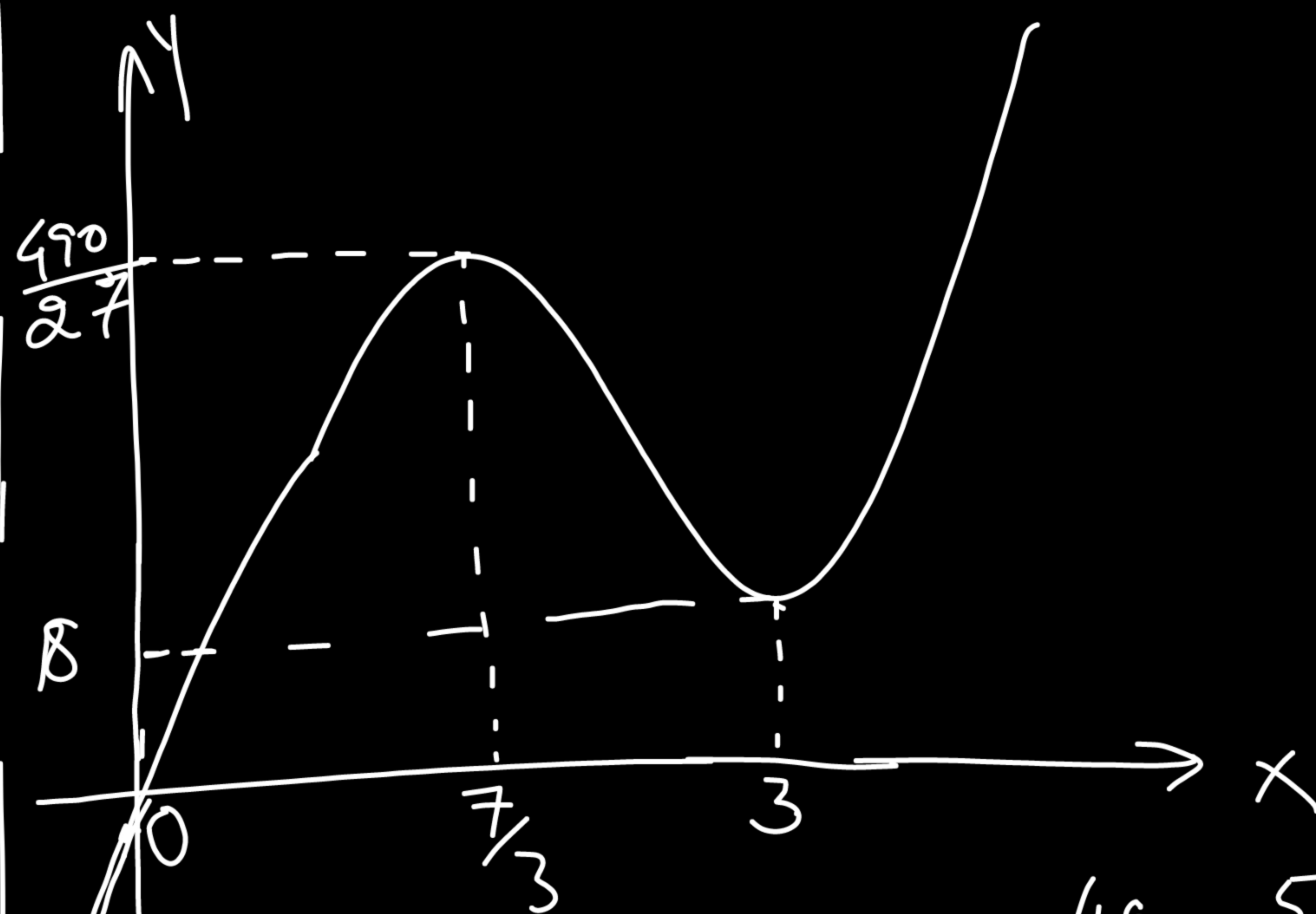
$$y' = 3t^2 - 16t + 21 = 0$$

$$t = \frac{16 \pm \sqrt{256 - 252}}{6}$$

$$= \frac{16 \pm 2}{6} = \frac{8 \pm 1}{3}$$

$$= 3, \frac{7}{3}$$

$$\frac{84}{252}$$



$$y = t(t^2 - 8t + 21)$$

$$y = 3(9 - 24 + 21) = 18$$

$$y = \frac{7}{3} \left(\frac{49}{9} - \frac{56}{3} + 21 \right) = \frac{7}{3} \left(\frac{49 - 168 + 189}{9} \right) = \frac{490}{27}$$

$$\begin{array}{r} 189 \\ 49 \\ \hline 238 \\ 168 \\ \hline 70 \end{array}$$